

Combinational Circuits - I

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Agenda

- ◆ Design of combinational circuits
- ◆ Delays in Combinational Circuits
- ◆ Adders & Subtractors

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Digital Logic Circuits

Classification of Logic Circuits

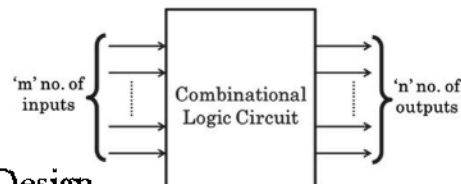
- ♦ **Combinational Logic Circuits**
 - ♦ Logic Circuits whose output is completely dependent on the combination of inputs.
- ♦ **Sequential Logic Circuits**
 - ♦ Logic Circuits whose output is dependent on the combination of inputs and past state of the circuit.
 - ♦ discussed in the later chapters.

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Combinational Logic Circuits

Block Diagram of Combinational Circuits



Steps to Design

- ♦ Understand Problem description and Draw the Block Diagram with I/P & O/P lines.
- ♦ Define the truth table based on the functionality.
- ♦ Get the simplified expression.
- ♦ Implement the Logic circuit.

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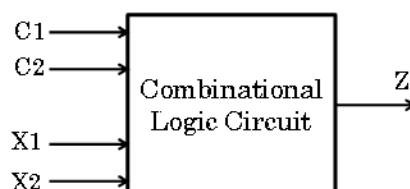
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Combinational Logic Design

Example : Design a combinational circuit with the specifications given below.

A switching circuit has two control inputs (C1,C2),two data inputs (X1,X2) and one output (Z). The circuit performs one of the logic operations on the two data inputs. The function performed depends on the control inputs as shown below



C1	C2	Operation
0	0	OR
0	1	EXOR
1	0	AND
1	1	EQU

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Truth Table

C1	C2	Operation
0	0	OR
0	1	EXOR
1	0	AND
1	1	EQU

With 4 input lines, 16 combinations of i/p's are possible.

C1	C2	X1	X2	Output (Z)	
0	0	0	0	0	OR
0	0	0	1	1	
0	0	1	0	1	
0	0	1	1	1	
0	1	0	0	0	EXOR
0	1	0	1	1	
0	1	1	0	1	
0	1	1	1	0	
1	0	0	0	0	AND
1	0	0	1	0	
1	0	1	0	0	
1	0	1	1	1	
1	1	0	0	1	EXNOR
1	1	0	1	0	
1	1	1	0	0	
1	1	1	1	1	

$$f(C1, C2, X1, X2) = \sum m(1, 2, 3, 5, 6, 11, 12, 15)$$

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Minimization

$$f(C1, C2, X1, X2) = \sum m(1, 2, 3, 5, 6, 11, 12, 15)$$

		X1, X2			
		00	01	11	10
C1, C2	00	0	1	1	1
	01	0	1	0	1
	11	1	0	1	0
	10	0	0	1	0

Possible groupings:

$$Z = C1'X1'X2 + C1'X1X2' + C1X1X2 + C1'C2'X2 + C1C2X1'X2'$$

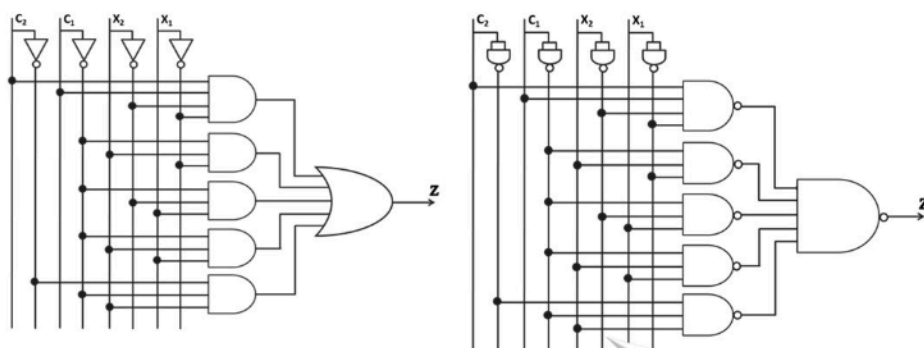
$$Z = C1'X1'X2 + C1'X1X2' + C1C2X1'X2' + C1X1X2 + C1'C2'X1$$

$$Z = C1'X1'X2 + C1'X1X2' + C1C2X1'X2' + C1X1X2 + C2'X1X2$$

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Implementation

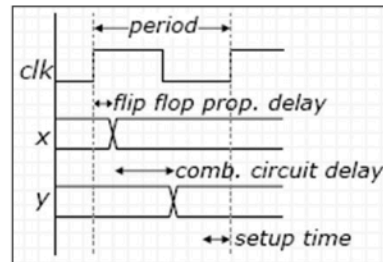
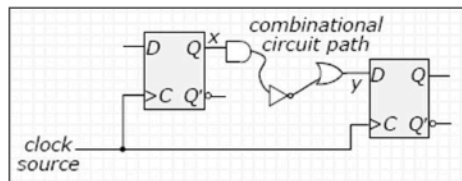


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RTL Design



$$\text{Clock Period } T = T_{\text{hold}} + T_{\text{pd}} + T_{\text{setup}}$$

$$\text{Frequency} = 1 / T$$

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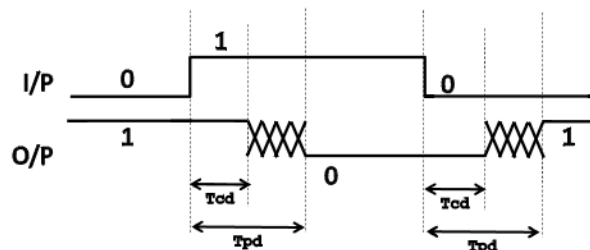
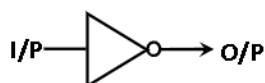
Delays in Combinational Circuits

Contamination Delay (T_{cd})

The amount of time needed for a change in a logic input to result in an initial change at an output.

Propagation Delay (T_{pd})

The amount of time needed for a change in a logic input to result in a permanent change at an output.



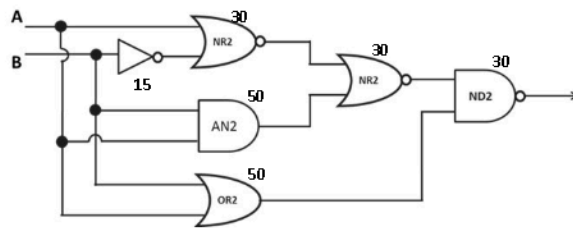
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Delays in Combinational Circuits

Example : Using the following table of timing specifications for each component, what is the TPD value for the circuit shown below?



Gate	T_{cd}	T_{pd}
INV	3ps	15ps
ND2	5ps	30ps
AN2	12ps	50ps
NR2	5ps	30ps
OR2	12ps	50ps

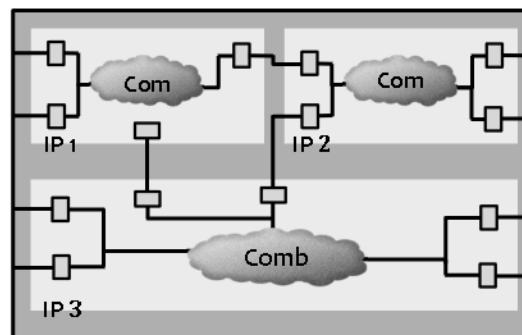
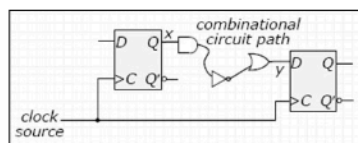
T_{pd}	90	105	110	80
	NR2, NR2, ND2	INV, NR2, NR2, ND2	AN2, NR2, ND2	OR2, ND2

110 ps is the maximum delay. So, the T_{pd} of this circuit is 110ps.

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RTL Design



$$\text{Clock Period } T = T_{\text{hold}} + T_{pd} + T_{\text{setup}}$$

$$\text{Frequency} = 1 / T$$

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Adders & Subtractors

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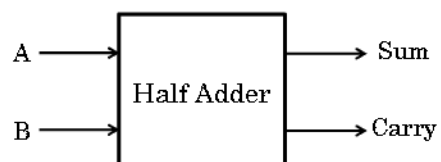
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Adders – Half Adders

Specifications of Half Adder

Adds 2 bits - A,B & produces Sum & Carry as Outputs

Block Diagram, Truth Table, Boolean Expressions



A	B	Carry	Sum
0	0	0	0
0	1	0	1
1	0	0	1
1	1	1	0

$$Sum = \bar{A}B + A\bar{B} = A \oplus B$$

$$Carry = AB$$

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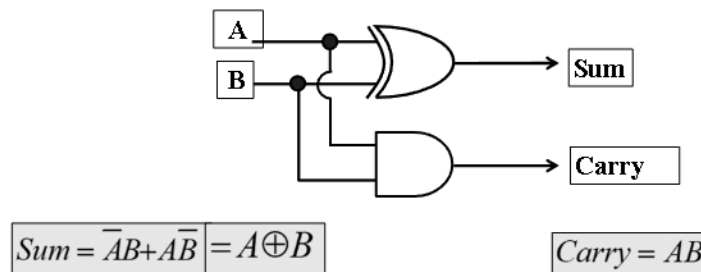
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Adders – Half Adders

Specifications of Half Adder

Adds 2 bits - A,B & produces Sum & Carry as Outputs

Circuit Diagram



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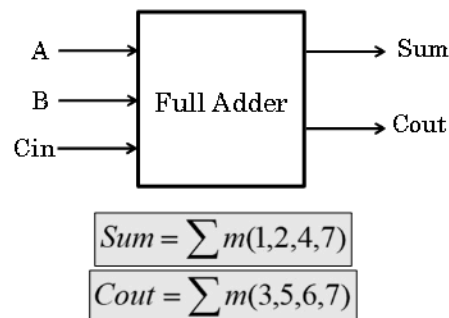
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Adders – Full Adders

Specifications of Full Adder

Adds 3 bits - A,B & Carry-in and produces Sum & Carry-out as Outputs

Block Diagram, Truth Table, Boolean Expressions



A	B	Cin	Cout	Sum
0	0	0	0	0
0	0	1	0	1
0	1	0	0	1
0	1	1	1	0
1	0	0	0	1
1	0	1	1	0
1	1	0	1	0
1	1	1	1	1

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Implementation of Full Adder

Boolean Expressions

$$Sum = \sum m(1,2,4,7)$$

A \ BC	BC			
	00	01	11	10
0	0	1	0	1
1	1	0	1	0

$$Sum = AB'C + A'B'C + ABC + A'BC'$$

$$Sum = A \oplus B \oplus C$$

$$Cout = \sum m(3,5,6,7)$$

A \ BC	BC			
	00	01	11	10
0	0	0	1	0
1	0	1	1	1

$$Cout = AB + BC + AC$$

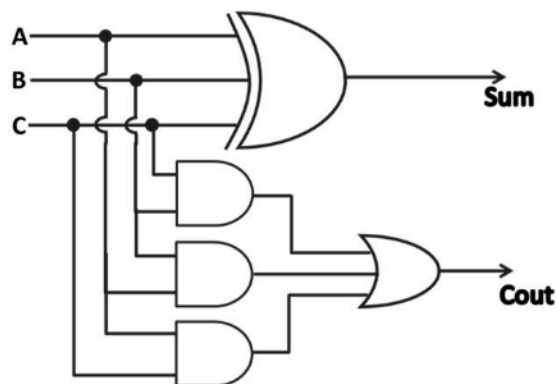
$$Cout = AB + BC + AC$$

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Implementation of Full Adder

Logic Circuit



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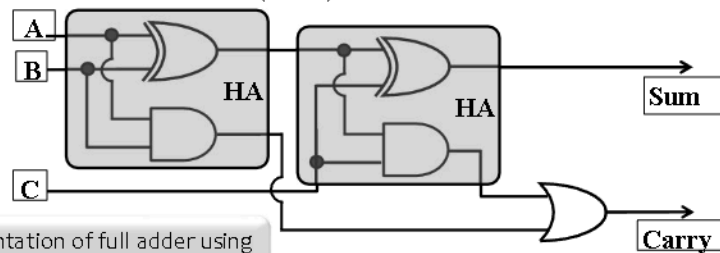
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Another Version

$$Sum = A \oplus B \oplus C$$

$$Cout = AB + BC + AC$$

$$\begin{aligned} Cout &= A.B + B.C + A.C \\ &= AB + A'BC + ABC + AB'C + ABC \\ &= AB(1 + C + C) + C(A'B + AB') \\ &= AB + C(A \oplus B) \end{aligned}$$



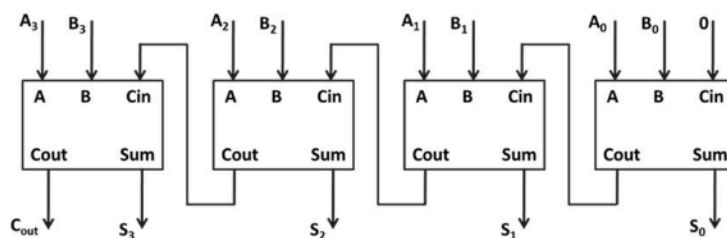
Implementation of full adder using
Two half adders & OR Gate.

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4-bit Adder

		1	1	0				C_2	C_1	C_0		
10	→	1	0	1	0	A	→	A_3	A_2	A_1	A_0	
+ 7	→	+	0	1	1	1	+ B	→	B_3	B_2	B_1	B_0
17	←	1	0	0	0	1		C_{out}	S_3	S_2	S_1	S_0

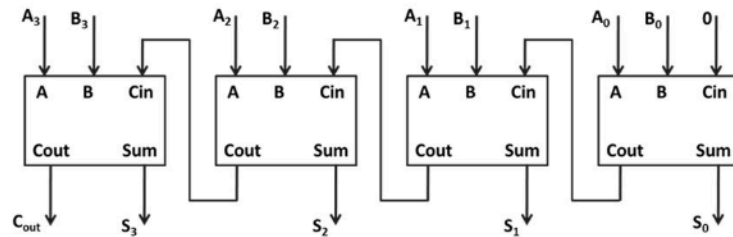


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Delays in a 4-bit Adder

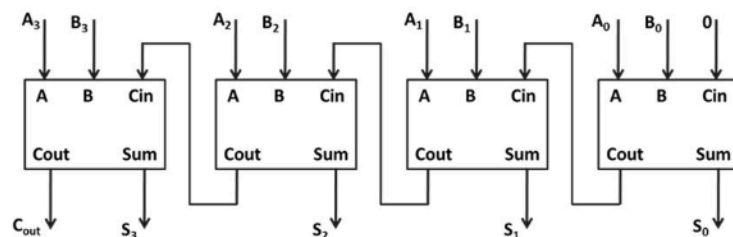


- Carry is propagated as a ripple from one stage to other stage. So, this adder is called as “ripple carry adder”
- In this adder, the addition process in each stage is waiting for the carry to be arrived from the previous stage of addition.
- So, the propagation delay of this adder is 4 times the Tpd of each adder. Hence, this adder is slow.

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Delays in a 4-bit Adder



- To make the addition process much faster, we need to make the process of addition independent on the previous stage addition.
- Carry Look Ahead Adder is an adder in which, addition process in each stage is carried out without having a dependency on the previous stage.

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4-bit Carry Look Ahead Adder

Truth Table of Full Adder

A	B	Cin	C-out	Sum
0	0	0	0	0
0	0	1	0	1
0	1	0	0	1
0	1	1	1	0
1	0	0	0	1
1	0	1	1	0
1	1	0	1	0
1	1	1	1	1

$$P = A \oplus B$$

$$G = AB$$

A + B generates carry, if and only if both A and B are 1

A + B propagates carry, if either one of A or B is 1

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4-bit Carry Look Ahead Adder

$$P = A \oplus B$$

$$G = AB$$

$$S_i = A_i \oplus B_i \oplus C_{i-1}$$

$$C_i = A_i B_i + C_{i-1} (A_i \oplus B_i)$$

$$S_i = P_i \oplus C_{i-1}$$

$$C_i = G_i + C_{i-1} P_i$$

$$S_0 = P_0 \oplus C_{-1} \Rightarrow S_0 = P_0$$

$$C_0 = G_0 + C_{-1} P_0 \Rightarrow C_0 = G_0$$

$$S_1 = P_1 \oplus C_0 \Rightarrow S_1 = P_1 \oplus G_0$$

$$C_1 = G_1 + C_0 P_1 \Rightarrow C_1 = G_1 + G_0 P_1$$

$$S_2 = P_2 \oplus C_1$$

$$C_2 = G_2 + C_1 P_2$$

$$\Rightarrow S_2 = P_2 \oplus (G_1 + G_0 P_1)$$

$$\Rightarrow C_2 = G_2 + (G_1 + G_0 P_1) P_2$$

$$S_3 = P_3 \oplus C_2$$

$$C_3 = G_3 + C_2 P_3$$

$$\Rightarrow S_3 = P_3 \oplus (G_2 + (G_1 + G_0 P_1) P_2)$$

$$\Rightarrow C_3 = G_3 + (G_2 + (G_1 + G_0 P_1) P_2) P_3$$

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4-bit Carry Look Ahead Adder

$$P = A \oplus B \quad G = AB$$

$$S_i = A_i \oplus B_i \oplus C_{i-1}$$

$$S_i = P_i \oplus C_{i-1}$$

$$\Rightarrow S_0 = P_0$$

$$\Rightarrow S_1 = P_1 \oplus G_0$$

$$\Rightarrow S_2 = P_2 \oplus (G_1 + G_0 P_1)$$

$$\Rightarrow S_3 = P_3 \oplus (G_2 + (G_1 + G_0 P_1) P_2)$$

All sum & carry expressions are independent of any carry from the previous stage.

$$C_i = A_i B_i + C_{i-1} (A_i \oplus B_i)$$

$$C_i = G_i + C_{i-1} P_i$$

$$\Rightarrow C_0 = G_0$$

$$\Rightarrow C_1 = G_1 + G_0 P_1$$

$$\Rightarrow C_2 = G_2 + (G_1 + G_0 P_1) P_2$$

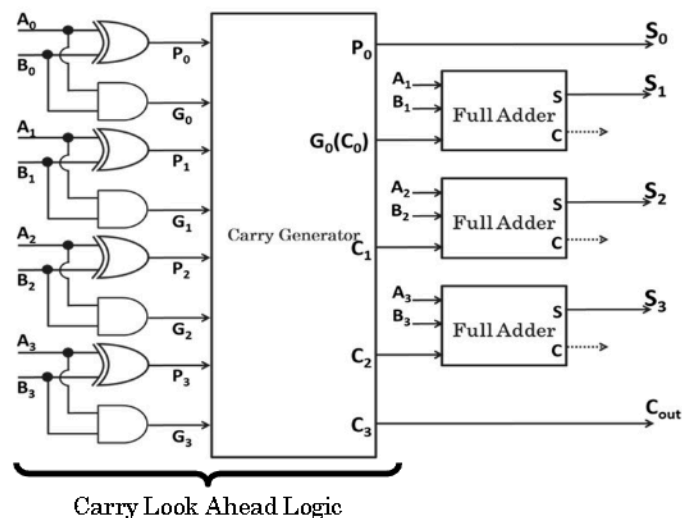
$$\Rightarrow C_3 = G_3 + (G_2 + (G_1 + G_0 P_1) P_2) P_3$$

So, these expressions can be used to realize the hardware of the adder to improve the speed.

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4-bit Carry Look Ahead Adder



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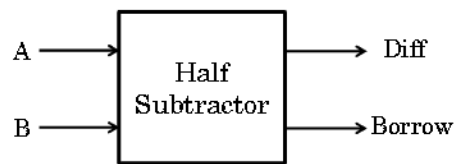
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Subtractors – Half Subtractor

Specifications of Half Subtractor

Subtracts B from A & produces Difference & Borrow as Outputs

Block Diagram, Truth Table, Boolean Expressions



A	B	Borrow	Diff
0	0	0	0
0	1	1	1
1	0	0	1
1	1	0	0

$$Diff = \bar{A}B + A\bar{B} = A \oplus B$$

$$Borrow = \bar{A}B$$

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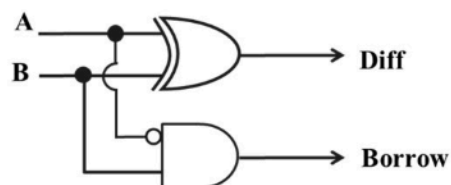
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Subtractors – Half Subtractor

Specifications of Half Subtractor

Subtracts B from A & produces Difference & Borrow as Outputs

Circuit Diagram



$$Diff = \bar{A}B + A\bar{B} = A \oplus B$$

$$Borrow = \bar{A}B$$

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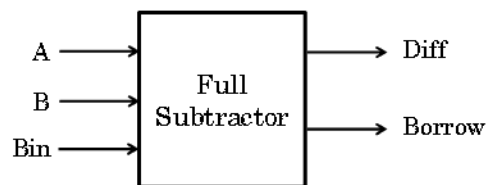
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Adders – Full Subtractor

Specifications of Full Subtractor

Subtracts B and Borrow-in from A

Block Diagram, Truth Table, Boolean Expressions



$$Diff = \sum m(1,2,4,7)$$

$$Borrow = \sum m(1,2,3,7)$$

A	B	B-in (C)	Borrow	Diff
0	0	0	0	0
0	0	1	1	1
0	1	0	1	1
0	1	1	1	0
1	0	0	0	1
1	0	1	0	0
1	1	0	0	0
1	1	1	1	1

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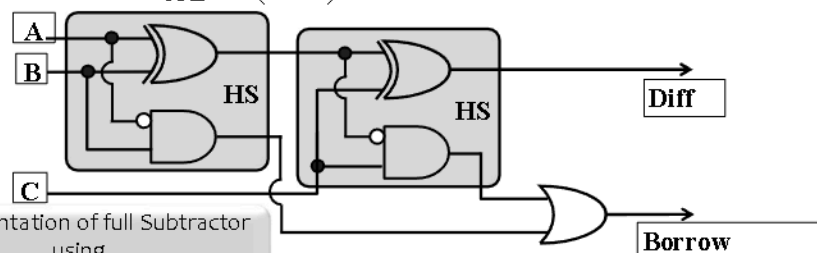
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Another Version

$$Diff = A \oplus B \oplus C$$

$$Borrow = \bar{A}B + BC + \bar{A}C$$

$$\begin{aligned}
 Borrow &= A'B + A'C + B.C \\
 &= A'B + A'BC + A'B'C + A'BC + ABC \\
 &= A'B(1 + C + C) + C(A'B' + AB) \\
 &= A'B + C(A \wedge B)'
 \end{aligned}$$



Implementation of full Subtractor using Two half subtractors & OR Gate.

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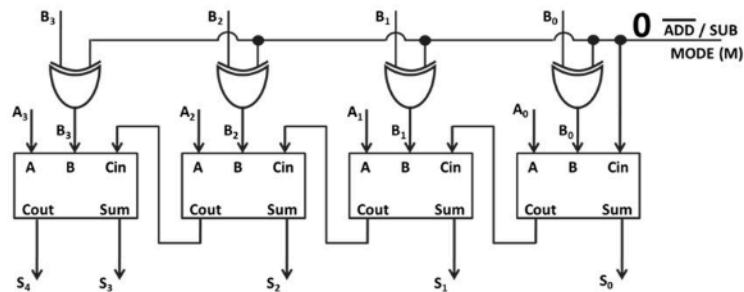
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Adder Subtractor Circuit

$$A + B \Rightarrow A + B \Rightarrow A + B + 0$$

$$A - B \Rightarrow A + 2's \text{ Comp}(B) \Rightarrow A + 1's \text{ Comp}(B) + 1$$



$$\text{Output} = A + B + 0 = \text{Sum}$$

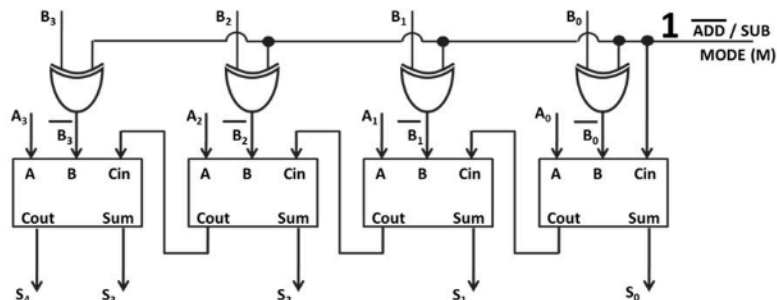
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Adder Subtractor Circuit

$$A + B \Rightarrow A + B \Rightarrow A + B + 0$$

$$A - B \Rightarrow A + 2's \text{ Comp}(B) \Rightarrow A + 1's \text{ Comp}(B) + 1$$



$$\text{Output} = A + \overline{B} + 1 = \text{Diff}$$

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